Newman-Penrose Formalism Calculations

Stephani et al., Equation (13.19_B0)

Metric

$$ds^2 = -A\left(w\right)^2 e^y \mathrm{d}v \otimes \mathrm{d}u - A\left(w\right)^2 e^y \mathrm{d}u \otimes \mathrm{d}v + \mathrm{d}w \otimes \mathrm{d}w + A\left(w\right)^2 \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$$\left(x^\mu\right) = \left(v, u, w, y\right)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{2}A\left(w\right)^2\right)\, dv$$

$$n = \left(\frac{1}{2}\sqrt{2}e^y\right)\, du$$

$$m = \left(\frac{1}{2}\sqrt{2}\right)\, dw + \left(\frac{1}{2}i\sqrt{2}A\left(w\right)\right)\, dy$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}\right)\, dw + \left(-\frac{1}{2}i\sqrt{2}A\left(w\right)\right)\, dy$$

Contravariant components

$$l = \left(-\sqrt{2}e^{(-y)}\right)\, du$$

$$n = \left(-\frac{\sqrt{2}}{2\,A\left(w\right)^{2}}\right)\,dv$$

$$m = \left(\frac{1}{2}\,\sqrt{2}\right)\,dw + \left(\frac{i\,\sqrt{2}}{2\,A\left(w\right)}\right)\,dy$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}\right)\,dw + \left(-\frac{i\,\sqrt{2}}{2\,A\left(w\right)}\right)\,dy$$

Directional Derivatives

$$D = -\sqrt{2}e^{(-y)}\,\partial_u$$

$$\Delta = -\frac{\sqrt{2}}{2\,A\left(w\right)^{2}}\,\partial_v$$

$$\delta = \frac{1}{2}\,\sqrt{2}\,\partial_w + \frac{i\,\sqrt{2}}{2\,A\left(w\right)}\,\partial_y$$

$$\overline{\delta} = \frac{1}{2}\,\sqrt{2}\,\partial_w - \frac{i\,\sqrt{2}}{2\,A\left(w\right)}\,\partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{i\sqrt{2}}{8\,A\,(w)}$$

$$\beta = \frac{\sqrt{2}\left(4\,\frac{\partial}{\partial w}A\,(w) - i\right)}{8\,A\,(w)}$$

$$\tau = -\frac{\sqrt{2}\left(2\,\frac{\partial}{\partial w}A\,(w) + i\right)}{4\,A\,(w)}$$

$$\pi = \frac{\sqrt{2}\left(2\,\frac{\partial}{\partial w}A\,(w) - i\right)}{4\,A\,(w)}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{4\,\frac{\partial}{\partial w}A\,(w)^2 - 4\,A\,(w)\,\frac{\partial^2}{(\partial w)^2}A\,(w) + 1}{8\,A\,(w)^2}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{4\,\frac{\partial}{\partial w} A\left(w\right)^2 - 4\,A\left(w\right)\,\frac{\partial^2}{(\partial w)^2} A\left(w\right) + 1}{16\,A\left(w\right)^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{4\,\frac{\partial}{\partial w} A\left(w\right)^2 - 4\,A\left(w\right)\,\frac{\partial^2}{(\partial w)^2} A\left(w\right) + 1}{8\,A\left(w\right)^2}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = -\frac{4\,\frac{\partial}{\partial w} A\left(w\right)^2 + 4\,A\left(w\right)\,\frac{\partial^2}{(\partial w)^2} A\left(w\right) + 1}{16\,A\left(w\right)^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"